

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

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Course Title : Software Development Fundamental-II
Course Code: 15B11CI211

Maximum Time: 1 Hour
Maximum Marks : 20

CO1	Explain various object-oriented concepts like class and objects, constructors, abstraction, data hiding (access specifics) etc.
CO2	Apply object-oriented concepts like inheritance, polymorphism and template to various real-world problems.
CO3	Apply SQL commands to create tables and perform various operations like insert, delete, select on tables.
CO4	Analyze the source code for the possible outcomes, exceptions and to debug the errors.
CO5	Design and implement the class diagrams for varied real life problem using OPPs.

Note: Attempt all the questions.

Q1. Create a class student having data members as name (string) and rollno (int). Each student can be either a cricketer or footballer or both. Every cricketer has runs (int) and wickets (int) while every footballer has goals scored (int) and assists (int). A star performer is a student who plays both cricket and football have a star performer index calculated using the following formula: $\text{star performer index} = \frac{[(\text{runs} - 1000) / \text{runs} + (\text{goal} - 10) / \text{goal}]}{2}$. Develop a C++ program that can simulate the following rules adhering to the above inheritance structure (you can define additional necessary method). [CO2(Applying), 5M]

- Diamond rule*
- Create a star performer object.
 - Set the student details and the cricketer/footballer or both by calling the appropriate methods from the star performer class.
 - Compute the star performer index value using the above mentioned formula.
 - Display the details of the students if he/she is a star performer. A Student is considered as a star performer if and only if star performance index is greater than zero.

Q2. Draw a UML class diagram for a restaurant, which has its name, location, owner name. Restaurant has traditional south Indian menu which is having dishes (items) details and prices, some are available all day and some are specifically available in breakfast and evening. Whenever customer with name, location and contact number, places any order, it contains the items from the menu and also their quantity and bill (total amount to pay). After the payment, the receipt is generated for the order having all food details and customer details. The restaurant has two types of staff members, chef staff for cooking and other operational staff for taking order and delivering it. A feedback is also taken from the customer after delivering his order. Show the complete diagram with appropriate relations with class details, types of relationship and its multiplicity.

UML [CO5(Creating), 5M]

Q3. An International ship can be classified as either a Cruise ship or a Cargo carrier. Each ship has a name (String), fuel capacity (int), valuation (double), owning company (string) and insurance premium (double). A cruise ship has additional features like number of passenger (int), number of pools (int), food storage (double) while a cargo carrier has number of crews (int), number of risky zones (int) and inflammable material quantity (double). The insurance premium for each type of ship is computed using following formula.

For cruise ship: $\text{insurance premium} = (\text{valuation} + \text{fuel capacity} + \text{food storage}) / (\text{number of passenger} * \text{number of pools})$

For cargo carrier: insurance premium = [(valuation + flue capacity) * number of risky zones + inflammable material quantity] / (number of passenger * 1000).

cruise

abstract classes

Design a C++ program that can simulate above scenario. Ensure that you use a computeInsurance() method to display the insurance premium and implement it in such a way so that it become mandatory to define it in all the cruise and cargo classes.

[CO2(Creating), 4M]

Q4. Do not consider syntax errors.

exception handling

(a) Predict the output and modify the program in so that it can print "Dino cannot be zero."
"Memory was allocated successfully but dino was zero!!".
[CO4(Analyzing), 3M]

```
#include<iostream>
using namespace std;
class Fraction
{ int num, dino;
public:
Fraction(){}
void setFraction(int n, int d){
if(d==0)
throw " Dino cannot be zero.";
else
num=n;
dino=d;
}};
int main()
{
int size=1, a=5, b=0;
Fraction *arr;
cin>>size;
arr = new Fraction[size];
try{
if(arr==NULL)
throw " Memory Error";
else{
for (int i=0; i<size; i++)
{
try{
arr[i].setFraction(a, b);
}
catch(const char * s){
count<<s<<endl;
throw s;
}}
}
}
catch(const char *s) {
count<<s<<endl;
}
return 0;}
```

(b) Predict the output and give suitable interpretation.
[CO4(Analyzing), 1.5M]

```
#include<iostream>
using namespace std;
int main()
{
char var= 'c';
try
{
try {
cout<<"inside try block\n";
if(var>0)
{
throw var;
cout <<"After throw\n"
}
}
catch(int var) {
cout<<"Exception Caught\n";
throw(var);
}
}
catch( char var)
{
cout<<"Exception Recaught\n";
cout<<"After catch\n";
return 0;
}
```

(c) Predict the output and give suitable interpretation.
[CO4(Analyzing), 1.5M]

```
#include<iostream>
#include<string>
using namespace std;
class base
{
public:
void display (int x) {
cout<<"Base Class"<<endl;
};
};
class Derived: public base {
public: void display()
{
cout << Derived Class"<<endl; }
};
int main ()
{
Derived D;
D.display(5);
return 0;
}
```

Inheritance